Ref.	Author & Venue	Paper Title & Publication Date (Vol/Issue/Pages)	Core AI Technique	Target Problem	Key Limitation / Research Gap
[1]	Marcus, R., et al. (PVLDB)	"Neo: A Learned Query Optimizer" (2019)	Deep Reinforcement Learning (DQN)	Learning optimal join orderings.	Severe "coldstart" problem; requires millions of queries to train.
		PVLDB, Vol. 12, Issue 11, Pages 1705-1718.			
[2]	Hilprecht, B., et al. (PVLDB)	"DeepDB: Learn from Data, not from Queries!" (2020)	Sum-Product Networks (SPNs)	Learned cardinality estimation.	SPN training scales poorly with highdimensional, correlated datasets.
		PVLDB, Vol. 13, Issue 7, Pages 992-1005.			
[3]	Ortiz, J., et al. (PVLDB)	"DQ: Learning to Optimize Join Queries With DRL" (2019)	Deep QNetworks (DQN)	Join ordering in complex query graphs.	Action space explodes for >10 joins; struggles with cyclic graphs.
		PVLDB, Vol. 12, Issue 11, Pages 1451-1464.			
[4]	Zhu, R., et al. (PVLDB)	"Lero: A Learningto-Rank Query Optimizer" (2023)	Learning-toRank (LTR)	Continuously improving native query optimizers.	Conservative bounds cause frequent fallbacks to traditional optimizers.
		PVLDB, Vol. 16, Issue 6, Pages 1466-1479.			
[5]	Han, S., et al. (PVLDB)	"LEON: A Learned Query Optimizer for Logical Plan Generation" (2022)	Sequence-to-Sequence (Transformer)	Generating logical query plans from SQL.	[ALCD] Generates syntactically invalid plans due to lack of CFG constraints.
		PVLDB, Vol. 15, Issue 11, Pages 2839-2852.			

[6]	Yu, X., et al. (PVLDB)	"Cost-based or Learning-based? A Hybrid Query Optimizer" (2022)	Hybrid (Traditional + ML)	Combining cost-based and learning-based optimization.	Still relies heavily on traditional optimizer fallbacks for complex queries.
		PVLDB, Vol. 15, Issue 11, Pages 3924-3937.			
[7]	Li, G., et al. (PVLDB)	"QTune: A Query-Aware Database Tuning System with DRL" (2019)	Deep Deterministic Policy Gradient (DDPG)	Recommending index configurations and memory knobs.	Continuous action space is unstable and highly workloadspecific.
		PVLDB, Vol. 12, Issue 12, Pages 2118-2130.			

[8]	Li, G., et al. (SIGMOD Record)	"AI Meets Database: AI4DB and DB4AI" (2021)	Survey / Transfer Learning concepts	Transferring optimizer knowledge across systems.	Negative transfer occurs if source and target database schemas are too dissimilar.
		SIGMOD Record, Vol. 50, Issue 2, Pages 5-16.			
[9]	Heinrich, R., et al. (PVLDB)	"Opening The Black-Box: Explaining Learned Cost Models" (2025)	Explainable AI (XAI) / SHAP	Explaining why an AI chose a specific plan or index.	Post-hoc explanations are approximations; lack formal mathematical guarantees.
		PVLDB, Vol. 18.			
[10]	Marcus, R., & Papaemmanouil, O. (PVLDB)	"Plan-structured deep neural network models for query performance prediction" (2019)	Plan-structured Neural Networks (Tree-CNNs)	Predicting query execution latency to guide the optimizer without executing the query.	Relies heavily on historical execution traces for training. [ALCD Connection] It models the plan statically and ignores the low-level, compiler-style optimizations (like Basic Block

					DAGs) that dictate actual CPU costs on lightweight, local hardware.
		PVLDB, Vol. 12, Issue 11, Pages 2033-2046.			